LU

ex w/ steps:

$$\begin{bmatrix} 2 & 5 & 2 \\ 6 & 5 & 3 \\ 1 & 3 & 1 \end{bmatrix}$$

1) $A = \begin{pmatrix} 2 & 5 & 2 \\ 6 & 5 & 3 \\ 1 & 3 & 1 \end{pmatrix}$ $L = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$U = \begin{pmatrix} 2 & 5 & 2 \\ 6 & 5 & 3 \\ 1 & 3 & 1 \end{pmatrix}$

2) $U[0,0] = \begin{bmatrix} 2 \\ 6 \\ 1 \end{bmatrix}$, turn 6 (Row 2, Col 1) into 0

$m_{21} = a_{21}/a_{11} = 6/2 = 3$

Row 2 $-$ (Row 1 $*$ m_{21}) $= (6, 5, 3) - (6, 15, 6)$

^{N}Row 2 $= (0, -10, -3)$

turn 1 (Row 3, Col 1) into 0

$m_{31} = a_{31}/a_{11} = 1/2$

Row 3 $-$ (Row 1 $*$ m_{31}) $= (1, 3, 1) - (1, 5/2, 1)$

^{N}Row 3 $= (0, 1/2, 0)$

$$U = \begin{pmatrix} 2 & 5 & 2 \\ 0 & -10 & -3 \\ 0 & 1/2 & 0 \end{pmatrix}, \quad L = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 1/2 & m_{32} & 1 \end{pmatrix}$$

$U[1,1] = \begin{bmatrix} 5 \\ -10 \\ 1/2 \end{bmatrix}$, turn 1/2 (Row 3, col 2) into 0

$m_{32} = a_{32}/a_{22} = \frac{1/2}{-10} = -.05$

Row 3 $-$ (Row 2 $*$ m_{32}) $= (0, 1/2, 0) - (0, 1/2, .15)$

^{N}Row 3 $= (0, 0, -.15)$

$$U = \begin{pmatrix} 2 & 5 & 2 \\ 0 & -10 & -3 \\ 0 & 0 & -.15 \end{pmatrix}, \quad L = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 1/2 & -.05 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 5 & 2 \\ 6 & 5 & 3 \\ 1 & 3 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 1/2 & -.05 & 1 \end{pmatrix} \begin{pmatrix} 2 & 5 & 2 \\ 0 & -10 & -3 \\ 0 & 0 & -.15 \end{pmatrix}$$

$P \rightarrow I$ w/ rows moved w/ permutations

check LU function:

$$A = \begin{bmatrix} 2 & 5 & 2 \\ 6 & 5 & 3 \\ 1 & 3 & 1 \end{bmatrix}$$

n = A.shape[0], 3

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad U = \begin{pmatrix} 2 & 5 & 2 \\ 6 & 5 & 3 \\ 1 & 3 & 1 \end{pmatrix}$$

num_rows = len(A) = 3
num_cols = len(A[0]) = 3

col	0	1	2
0	2	5	2
1	6	5	3
2	1	3	1
row			

for col in range(num_cols):
  for row in range(num_rows):
    if row == col:

* row = 0, col = 0 ✓

i = row, 0
k = row+1, 1
while k, 1 < U.shape[0], 3:
L[1][0] = U[1][0] / U[0][0]
L[1][0] = 6/2 = 3
multiplier = L[1][0], 3
pivot_row = U[i] = U[0]
↳ [2, 5, 2]
m_row = pivot_row * multiplier
↳ [2,5,2] * 3 = [6, 15, 6]
new_row = U[k], U[1] = m_row
↳ [6,5,3] − [6,15,6]
[0, −10, −3]
U[k], U[1] = new_row

$$U = \begin{pmatrix} 2 & 5 & 2 \\ 0 & -10 & -3 \\ 1 & 3 & 1 \end{pmatrix}, \quad L = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

k += 1 = 2 ↻

if 1 == 1:

* row = 1, col = 1

i = row, 1
k = row+1, 2
while 2 < 3:
L[2,1] = ½ / −10 = −.05
multiplier = −.05
pivot_row = [0, −10, −3]
m_row = [0, ½, .15]
new_row = [0, ½, 0] − [0, ½, .15]
new_row = [0, 0, −.15]
U[2] = [0, 0, −.15]

$$U = \begin{pmatrix} 2 & 5 & 2 \\ 0 & -10 & -3 \\ 0 & 0 & -.15 \end{pmatrix}, \quad L = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 1/2 & -.05 & 1 \end{pmatrix}$$

k += 1 = 3

while 3 < 3: → break

if 2 == 1: x

↻

while 2 < 3:
L[2][0] = ½
multiplier = ½
pivot_row = [2, 5, 2]
m_row = [1, 5/2, 1]
new_row = [1, 3, 1] − [1, 5/2, 1]
↓ [0, ½, 0]
U[2] = new_row

$$U = \begin{pmatrix} 2 & 5 & 2 \\ 0 & -10 & -3 \\ 0 & ½ & 0 \end{pmatrix}, \quad L = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ ½ & 0 & 1 \end{pmatrix}$$

k += 1 = 3 ↻

while 3 < 3: → break

if 1 == 0: x

if 2 == 0: x

⋮ ↻

if col == (num_cols − 1):
2 == 2
↳ return L, U